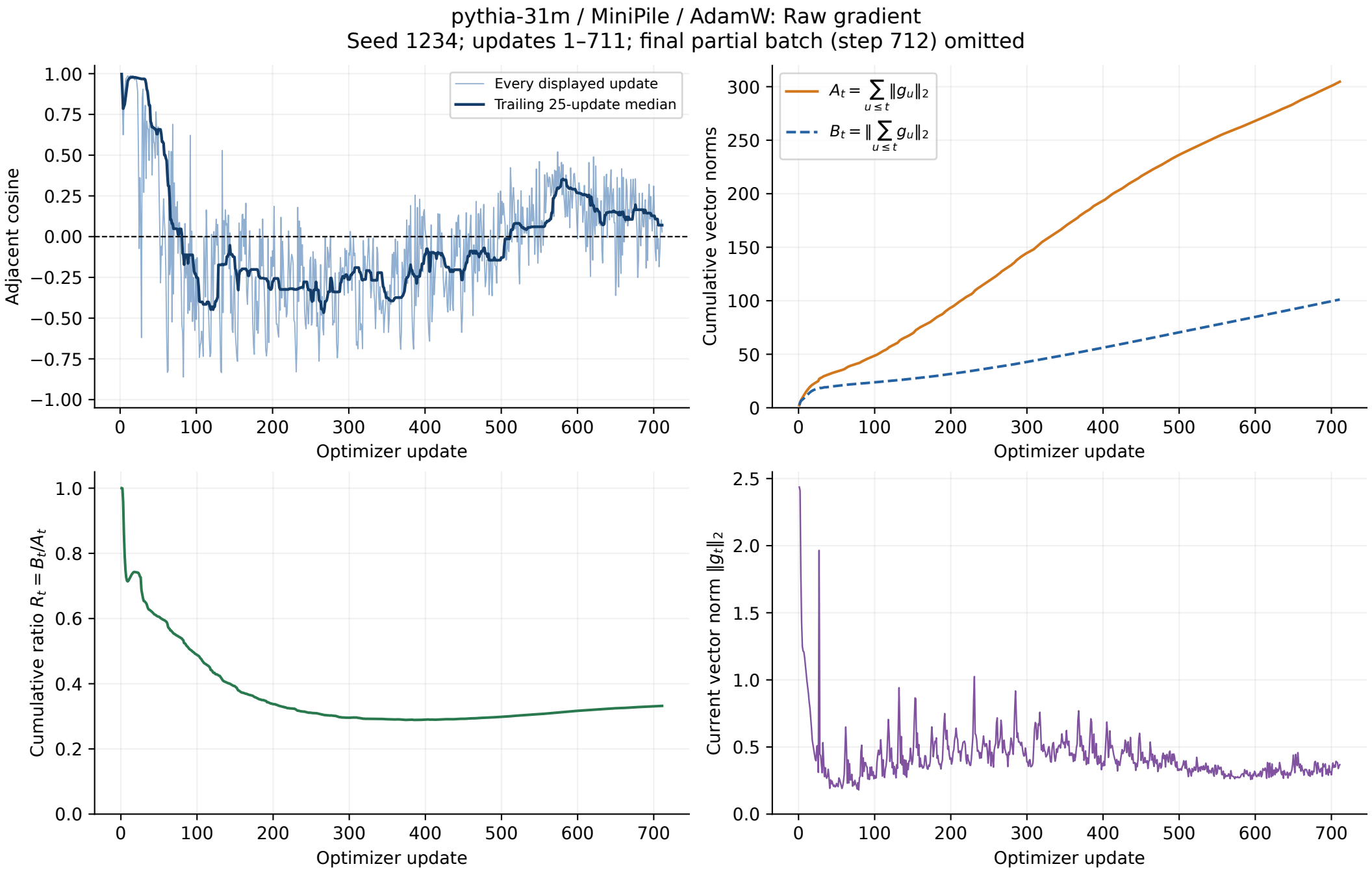

pythia-31m / MiniPile / AdamW: Raw gradient
Seed 1234; updates 1–711; final partial batch (step 712) omitted
Every displayed update
Trailing 25-update median
Adjacent cosine
Optimizer update
$A_t = \sum_{u \le t} \|g_u\|_2$
$B_t = \| \sum_{u \le t} g_u \|_2$
Cumulative vector norms
Optimizer update
Cumulative ratio $R_t = B_t / A_t$
Optimizer update
Current vector norm $\|g_t\|_2$
Optimizer update